

Project Design Phase-II

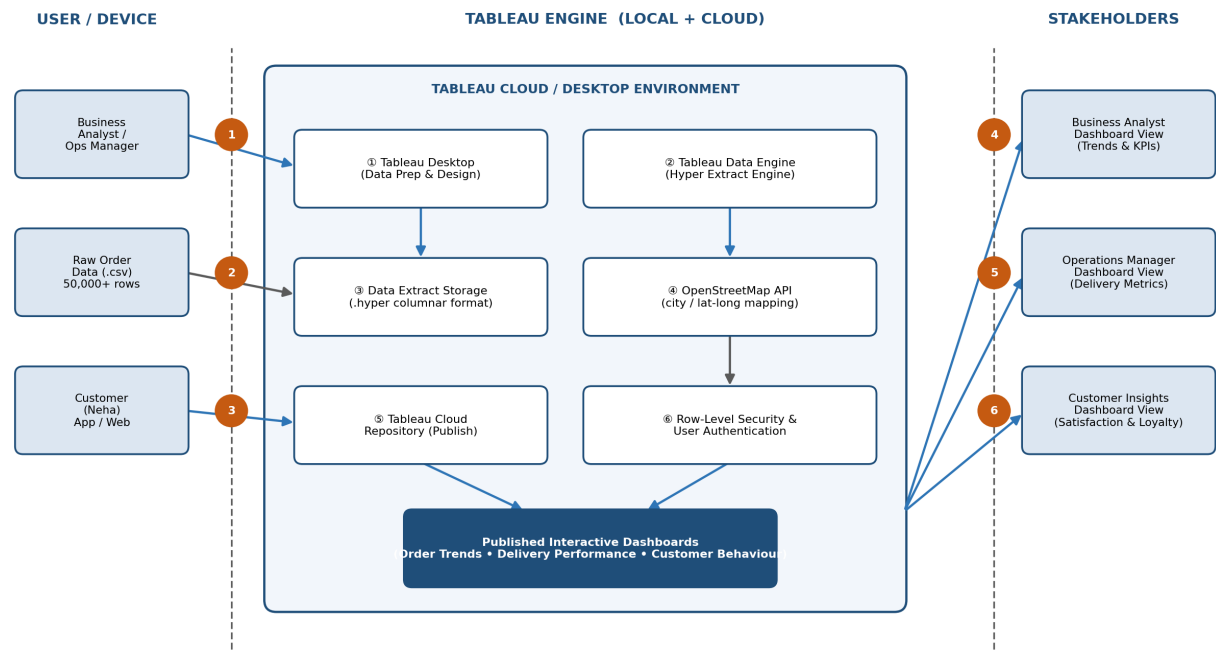
Technology Stack (Architecture & Stack)

Date	11 July 2026
Team ID	SWTID-2026-5277
Project Name	Food Ordering Behaviour and Consumer Trends
Maximum Marks	4 Marks

Technical Architecture:

The diagram below shows the end-to-end technical architecture used for this project, covering the three usage scenarios: **Rahul Sharma (Business Analyst)**, **Priya Mehta (Operations Manager)**, and **Neha Reddy (Customer)**. Raw order data flows from the local CSV source into the Tableau processing engine, where it is prepared, mapped, extracted and published as interactive dashboards for each stakeholder.

Scenario coverage: Order Trend & Cuisine Analysis, Delivery Performance Monitoring, and Customer Ordering Behaviour & Satisfaction Tracking, all powered by a single Tableau data pipeline sourced from the food ordering behaviour dataset.



Guidelines followed: application logic blocks shown (Tableau Desktop, Data Engine, Extract Storage), infrastructural demarcation between local (CSV / Desktop) and cloud (Tableau Cloud Repository) components, external interface indicated (OpenStreetMap API), and data storage components (.hyper extract format) clearly labelled.

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How operational management interacts with metrics and switches visual sheets.	Tableau Web Dashboard UI / Web Browser
2.	Application Logic-1	Core engine processing data parsing, table metrics, and interactive calculation fields.	Tableau Data Engine (Hyper Technology)
3.	Application Logic-2	Local workstation software used by team to design sheets, set colors, and map data fields.	Tableau Desktop
4.	Database	Local source storage holding raw records of customer orders and driver delivery times.	Flat Structured Data File (.csv format)
5.	Cloud Database	Web storage server utilised if publishing live or scaling workbook rows.	Tableau Cloud Server Repository
6.	File Storage	Local computer directory hosting the active dataset and workbook configurations.	Local NTFS / APFS Filesystem
7.	External API-1	Mapping coordination API embedded inside the engine to plot lat/long city boundaries.	OpenStreetMap API (Built-in to Tableau)
8.	Infrastructure (Server / Cloud)	Host server deployment environment allowing remote manager workspace access.	Tableau Cloud (SaaS) / Hosted Server

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Mention any open community or free-tier frameworks applied in mapping data elements.	OpenStreetMap spatial boundary metadata
2.	Security Implementations	Access controls to protect sensitive internal corporate logistics files from public leak.	Tableau Cloud User Authentication & Row-Level Permissions
3.	Scalable Architecture	Strategy to ensure extra monthly csv data uploads do not lag down dashboard performance.	Tableau Extract Columnar Storage (.hyper format)
4.	Availability	Ensuring the analytics dashboard is up whenever operational shifts require checking trends.	Tableau Cloud SaaS infrastructure offering 99.9% uptime
5.	Performance	Keeping visual updates crisp when clicking filters or loading data charts.	Pre-aggregated Data Extracts and In-memory Caching

References:

<https://c4model.com/>
<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
<https://www.ibm.com/cloud/architecture>
<https://aws.amazon.com/architecture>
https://help.tableau.com/current/pro/desktop/en-us/hyper_data_source.htm
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